

# **SURVIVAL ANALYSIS OF INITIAL PUBLIC OFFERINGS IN HOT AND COLD MARKET: AN ACCELERATED FAILURE TIME APPROACH**

**Dr. Garima Baluja**

(Corresponding Author)

Assistant Professor

DAV University, Jalandhar, Punjab

garima.baluja@gmail.com

**Dr. Balwinder Singh**

Associate Professor

Guru Nanak Dev University, Amritsar, Punjab.

bksaini@gmail.com

**Abstract:**

*The market conditions are most obvious and least controllable that influence the decision of a company to issue an IPO. The anecdotal literature suggests that timings are everything for an IPO. The market conditions are crucial not only for the performance of an issue but also for its survival on the trading exchange. The Indian primary market has witnessed several volatile market conditions in the post-SEBI era. The introduction of SEBI and abolition of CCI created 'hot issue phenomenon' in the market wherein several new issues entered the market, however, only a few managed to survive in the aftermarket. Such high failure rate of IPOs and diverse market conditions generated the need to explore the survival profile of IPOs in India. Using the most sophisticated methodologies i.e., Logistic Regression and Survival Analysis, the present study explores the survival profile of IPOs in hot market conditions of India. The empirical investigation reveals that most of the IPOs entered the market in hot issue period (1992-1996) but they failed to survive longer in the aftermarket. Overall, the Kaplan-Meier estimation exhibits a significant decline in survival rate and a growth in hazard rate during the first 50-60 months of listing. The analysis of market specific variables and survival profile of IPOs reveals that issues in the period of high IPO activity fails to sustain longer on the exchange. Apart from this, several offering and company specific factors also exhibit significant results. The findings of this study will have fruitful implication for the issuers, investors, regulators, and the entire capital market as they can evaluate the future prospects of IPOs and can take rational decisions accordingly.*

**Keywords:** Initial Public Offerings, Kaplan-Meier, Logistic regression, Survival Analysis.

**JEL Classification:** G24; G30; G32; G33.

## 1. INTRODUCTION

Over a period, the issue of corporate failure has become a matter of concern in economic as well as business area. 'Failure' simply refers to the inability of a firm to meet its desirable objectives and viewed as the opposite of success (Walter, 1957; Donaldson, 1962; Li and Lui, 2010). One such kind of failure is the delisting of issue from the market. Delisting is a traumatic event for both the firm as well as the shareholders (Li *et al.*, 2006). The failure of issue on the trading exchange may lead to bankruptcy, liquidation or momentous changes in the control of a firm and consequently results in huge losses to firm (Noor and Iskandar, 2012). Further, it hampers the interest of investors, creditors, and the economy at large. In the past few years, investors in IPOs had truly a bittersweet experience due to such failures in the aftermarket (Peristiani, 2003). Agarwal and Gort (2002) observed that roughly 5-10 percent of the firms in the US left the market over the span of a single year. Similarly, Fama and French (2004) reported a significant increase in the number of new listings on the NASDAQ during the period 1973 and 2001 followed by a sharp decline in survival rates as well. Apart from issuers and investors, the efficiency as well as the functioning of the entire market is highly influenced when an issue fails to survive on the exchange. Since, the survival of IPO holds huge importance not only for the issuer but also for the investors as well as the economy at large, the research efforts in this area suddenly got a thrust.

Out of several market related factors, 'Market level' has been examined as an essential determinant for IPO survival by researchers (Hensler *et al.*, 1997; Goot *et al.*, 2011). Using Log-logistic AFT model, Hensler *et al.* (1997) found a negative relationship between market level and IPO survival time, i.e. IPOs during peak market have shorter lives. This is in line with the theoretical arguments of Brown (1970) which assert that the failure rate of IPO firms is cyclical

in nature. This means that the groups of firms going public in unreceptive market periods have lower failure rates as compared to the groups of firms going public in receptive periods. Also, Goot *et al.* (2011) found that IPOs brought during the period of high market level have a low survival rate as well as lesser survival duration in the market. However, Yang and Sheu (2006) failed to obtain any significant influence for the same. Contrary to this, Zhao (2005) found a positive influence of market level on the likelihood of survival of IPOs. They support this positive influence by stating that when returns during a period are encouraging the investors are more optimistic towards the new issue which makes such issue more likely to be successful in future. Hence, a firm is more likely to go public when market conditions are favorable as it enhances the survival prospects of issue in the aftermarket. Thus, the empirical support for market level and IPO survival is not very wide and no unanimity has been found which calls for more intensive research to establish the conclusiveness of this relationship.

IPO activity, which represents the number of issues in the market, is another significant factor which has been widely investigated in the literature. Examining the impact of ‘hot issue phenomenon’ on IPO survivability, Demers and Joos, 2006; Kooli and Meknassi; 2007; Carpentier and Suret, 2008, Raju and Prabhudesai, 2012, found that issues during hot period are more likely to fail as compared to issues in cold period. Further, Boubakri *et al.* (2005) and Hamza and Kooli (2010) confirm the ‘window of opportunity’ hypothesis and empirically support that IPO market activity affects negatively and significantly the probabilities of surviving and being acquired relative to non-surviving. Apart from the likelihood of survival, the issuance during the bubble period significantly reduces survival time as well (Espanlaub *et al.*, 2009). Hence, fierce competition in the market due to high entrants negatively influences the endurance of IPOs in the aftermarket (Kauffman and Wang, 2003). However, Yang and Sheu, 2006; Adjei

*et al.*, 2008, Chancharat *et al.*, 2008, did not find this variable to be statistically significant in influencing the survival prospects of IPOs. Similarly, Ahmad (2012) observed that about 60 per cent of firms in the UK came up with their IPO in hot market and consequently such hot period IPOs become non-survivors in the aftermarket, however the survival analysis models exhibit no statistical significance for this variable. On the other hand, Hensler *et al.* (1997) observed that IPO activity enhances the survival time of IPOs in the aftermarket. They found that issues during the high activity period survives significantly longer than issues during low activity, but provide no theoretical support for the same. Similarly, Kauffman and Wang (2003) found that as the number of IPO firm increases, the odds of bankruptcy decreases. Hence, these contradictory findings call for substantial evidence to understand how the survivability of IPO responds to this factor.

The stock exchange on which an issue gets listed is another factor examined for IPO survival. Peristiani (2003) found that companies listed on the NYSE face stricter rules and hence are more likely to be dropped when they falter. Similarly, Yang and Ding (2012) examined the survival across two stock exchanges in China (Shanghai Stock Exchange and the Shenzhen Stock exchange) wherein one exchange is more rigorous than other. But the Cox proportional hazard model exhibited no significant influence of the stock exchange on IPO survival. Boulton *et al.* (2007) observed that IPOs listed on Nasdaq are more likely to remain independent in the long run. Hence, the empirical evidence on influence of stock exchange is meager and needs further investigation.

A few scholars have examined the influence of interest rates on the survival prospects of IPOs in the aftermarket. In order to control the influence of macro economy, Kauffman and Wang (2003) used six months US treasury bill interest rate and examined its influence on IPO survival. The

results of logistic regression model revealed this variable to be positively and significantly associated with failure of IPOs suggesting that bankruptcy increases when interest rates climb. Although, the economic indicators are significant in determining the success of the IPO in the aftermarket, yet they have not been properly explored.

Although the efforts have been started across the world to explore the status of IPOs in the aftermarket, yet the reasons and consequences of delisting of IPOs are quite less explored, especially in India. As far as Indian IPO market is concerned, Raju and Prabhudesai (2012) explored the high failure rate of IPOs in light of global credit crunch and the US recession in 2007-2008. However, the empirical evidences on the survival profile of IPOs in the post-SEBI volatile market conditions are quite scarce. Hence, the dearth of literature on the survival profile of IPOs in India opens a scope for more research contribution.

The Indian IPO market has experienced several structural changes in the post-SEBI (Securities and Exchange Board of India) era. The abolition of the Controller of Capital Issues (CCI), establishment of SEBI, introduction of free pricing mechanism, and increase in participation by Foreign Institutional Investors (FIIs) has brought a sea change in the entire IPO market. The effect of such changes is evident from the upward trend in the IPO market during the period 1992-1996 (see figure1). However, along with this, several malpractices, discretionary allotments, and fly-by-night operators also entered the market that disrupted the smooth functioning of this market. Moreover, the Southeast crisis and the Internet bubble burst generated the negative sentiments among the investors and decelerated the growth of this market significantly.

*‘Insert Figure 1 here’*

During such period, several new issues failed to maintain their identity, whereas a few managed to sustain their status on the exchange. This signifies that market timings play a vital role in sustaining an issue on the exchange. The anecdotal literature also suggests that timing is everything for IPO (Zhao, 2005), however, no empirical evidence is found for the influence of such timings and scenario on the survival of IPOs in India. Hence, efforts have been made in this paper to explore the survival profile of IPOs in the post-SEBI volatile condition.

The aim of this paper is to fill the gap in literature by identifying the extent to which the post-IPO outcome varies, along with the determinants of IPO survival in the aftermarket. The study addresses this issue from three main perspectives. Firstly, the present study explores the role of timings of issue (hot or cold) in determining the success of IPOs on the exchange. The ‘hot issue’ period refers to the duration in which a large number of issues enter the market, whereas cold period attracts less number of issues (Ibbotson and Jaffe, 1975; Jain and Kini, 1999; Loughran and Ritter, 2004; Demers and Joos, 2007; Carpentier and Suret, 2008; Kooli and Meknassi, 2007). Researchers assert that hot issue period gives an immense number of issues in the market, but such issues are of low quality. Mainly, such low quality firms enter the market just to take the benefit of favorable market conditions, but in reality, they do not have the capacity to withstand the rough market conditions due to which they fail to survive in the aftermarket (Hensler *et al.*, 1997; Demers and Joos, 2007). However, in cold periods, stronger firms are more likely to succeed with their IPOs (Boubakri *et al.*, 2005). This situation is also known as ‘Window of Opportunity’ hypothesis. The present study attempts to empirically examine whether the phenomenon of ‘Hot issue period’ or ‘Window of Opportunity’ hypothesis have any significant influence on the survival profile of IPOs in India. Secondly, this study applies the survival analysis methodology, which is a unique way of exploring the duration of IPOs in the

aftermarket. Thirdly, it tracks down the effect of covariates on the post-listing status of IPOs that help in analyzing the significance of each factor in underpinning the two post-IPO outcomes.

The present study offers a distinct contribution to IPO literature in general and survival in particular. Further, it contributes in the area of survival analysis that has not been widely applied in the field of finance. The findings will be of great use for the issuers as they can critically evaluate the factors that are crucial for their survival and can build their strategies for the issues that would ensure their long run endurance on the exchange. In this way, they can uncover the reasons that are actually responsible for the failure of IPOs that needs to be given special attention. Apart from issuers, investors can evaluate the market as well as other control variables in order to ensure that their decision to invest in an issue should turn out to be profitable in the aftermarket. In practical terms, the findings of this study can inform public policy decision makers who are concerned with regulating the market. In other words, the study would provide a base for the regulators and policy makers to update their laws and formulate such kind of policies that would not only create a lucrative and more sustained market but will also protect the interest of investors in the aftermarket. In nutshell, the significance of analyzing the role of market conditions on the success or failure of IPOs would be immensely fruitful for every associated party.

This article is organized into four main sections. Section 1 introduces the topic, discusses the problem, and presents the literature in the area of IPO survival, Section 2 presents the database and methodology, Section 3 discusses the empirical results, and finally section 4 summarizes and concludes this paper.



## 2. DATABASE AND METHODOLOGY

**2.1 Data and Sample selection:** in order to capture the impact of market conditions, the initial data consists of IPOs that entered the market in the post-SEBI and hot issue period i.e., from 1992 to 1996. For sample selection, the data for various variables i.e., Share prices, issue size, subscription, name of lead managers, NIC code, and year of incorporation must be available. These criteria resulted in 3125 IPOs that got listed on Bombay Stock Exchange (BSE) from 1992-1996 and they are analyzed till the end of 2011.

**2.2 Sources for data collection:** Data for the variables i.e., Issue size, issue price, times subscribed, and IPO activity have been compiled from Prime database and Capitaline database. Incorporation year of each IPO and their National Industrial Classification (NIC 2008) codes has been obtained from Prowess database maintained by CMIE (Centre for Monitoring Indian Economy Pvt. Ltd.) on the basis of which IPOs have been classified into 10 major industries. The market returns for underpricing and market level have been computed by taking the closing values of Sensex from the official website of BSE. The data for post-listing IPO status, date and reason for delisting has been taken from BSE and Moneycontrol websites.

**2.3 Measurement of Variables:** The study defines an IPO as ‘survivor’ if it continues to list on the stock exchange and ‘non-survivor’ if it delists from the stock exchange due to liquidation, permanent suspension, compulsion by SEBI or any other reason, except due to its merger or movement to another stock exchange (Hensler *et al.*, 1997; Rath, 2008; Bhattacharya *et al.*, 2011). In order to predict the trajectories following the IPO, three sets of variables concerning offering, market, and corporate specific characteristics are taken. Table 1 presents the measurements of these variables.

*'Insert Table 1 here'*

**2.4 Empirical specifications:** The empirical analysis involves two dimensions. The determinants of IPO survival are empirically examined using 'Logistic Regression model', whereas the survival time of IPOs is explored using 'Survival Analysis methodology'.

Logistic Regression is a family of discrete choice models in which the dependent variable is categorical and independent variables can be continuous as well as categorical (Field, 2005, p. 218). The aim of this model is to assess how well the set of independent variables predicts the occurrence of the categorical dependent variable. The probability function in logistic regression is as follows:

$$Li = \ln \left[ \frac{Pi}{1-Pi} \right]$$
$$= \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_n X_n + \epsilon_i$$

Here,  $Li$  is the log of odds ratio;  $Pi$  is the probability that  $Y_i=1$  (i.e., An IPO continues to list on the exchange), and  $(1-Pi)$  = probability that  $Y_i=0$  (i.e., An IPO delists from the exchange);  $\beta_0$  is the constant;  $\beta_1, \beta_2, \beta_3 \dots \beta_n$  are the coefficients to be estimated.

Although the logit model is capable of predicting whether the event will occur or not, yet it gives no idea about the timings of that event. In other words, it makes no distinction between the firms failing in six months and the firms failing after two years (Lowers *et al.*, 1999; Kooli and Meknassi, 2007; Raju and Prabhudesai, 2012). Hence, to overcome this problem, survival analysis methodology is best suited. The survival analysis models not only examine the occurrence of the event, but also consider the timing of such event (Mills, 2010). In addition, this methodology deals with the censored data as well as time series data. Since IPO market

possesses both these features, hence this methodology is quite fruitful (Hamza and Kooli, 2010; Raju and Prabhudesai, 2012).

There are two main functions in survival analysis i.e, survival function and hazard function. The survival function refers to the probability that an individual will continue to survive until the end of the study period (Kleinbaum and Klein, 2005, p. 9) and is written as follows:

$$S(t) = \Pr(T > t) = 1 - F(t)$$

Here,  $S(t)$  = cumulative survival rate;  $T$  = time until the firm experiences the event (trading months);  $t$  = study time period;  $F(t)$  = cumulative density function =  $\Pr(T \leq t)$

Whereas, the hazard function is the measures of conditional probability that an IPO is delisted instantaneously, given that it has survived up to time  $t$ . It is denoted as (Lee and Wang, 2003, p. 11):

$$h(t) = \lim_{\Delta t \rightarrow 0} \frac{\Pr(t \leq T < t + \Delta t | T \geq t)}{\Delta t} = \frac{f(t)}{1 - F(t)} = \frac{f(t)}{S(t)}$$

Here,  $f(t)$  is the probability function that is the product of survival and the hazard function:

$$f(t) = S(t) h(t)$$

The survival analysis model follows several distribution forms such as non-parametric, semi-parametric, and parametric. The suitability of all such forms has been tested and accordingly non-parametric 'Kaplan Meier Estimation' method and parametric 'Accelerated Failure Time' (AFT) model has been employed. The model is written in log-linear as follows (Bradburn *et al.*, 2003):

$$\text{Ln}(T) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_p X_p + \varepsilon$$

Here,  $\text{Ln}(T)$  is the log of survival time, which is the dependent variable;  $\beta_0$  is the constant;  $\beta_1, \beta_2, \dots$  are coefficients of the covariates;  $X_1, X_2, X_3, \dots, X_p$  are the covariates;  $\varepsilon$  is the error term.

### 3. EMPIRICAL RESULTS

The survival of fittest IPOs are analyzed using different methodologies in the following sections:

**3.1 Descriptive Statistics:** Firstly, in order to gain an insight into the basic features of IPOs that entered the market in the post-SEBI volatile market conditions, their descriptive statistics are analyzed and compared across survivors and non-survivors using independent sample t test and Wilcoxon Z test. The tests exhibit that survivors have a significantly lower market level as well as IPO activity than non survivors. Also, survivors exhibit significantly higher issue size, demand, lead manager's reputation, and age in the market. However, the issues that are ill-fitted or fail to survive in the market conditions have significantly higher underpricing, risk, and listing delay. Table 2 displays the results of this analysis.

*'Insert Table 2 here'*

**3.2 Survival Pattern of IPOs:** The survival probabilities of IPOs can be assessed non-parametrically from the observed survival time for censored as well as non-censored observations through 'Kaplan-Meier estimation Method' (Kaplan and Meier, 1958). This method gives an in-depth understanding of survival as well as hazard patterns of IPOs in the aftermarket. It generates table and plots of survival as well as hazard function for event history data (Garson, 2012).

The mean and median of survival time is presented in table 3. This indicates that 95% of the IPOs fit to survive in between the period of 105 to 112 months (approx) in the aftermarket. Although the average months are 108 but the median time of survival is 54 months (approx).

*'Insert Table 3 here'*

Further, this method obtains the information of survived as well as failed IPOs and constructs the survival and hazard function plots over time. Such plots are known as 'Kaplan-Meier curves' (KM) which are the series of horizontal steps of declining magnitude. The KM survival curve is shown in figure 2, which summarizes the entire survival pattern of IPOs that entered the market during 1992-1996 and they are tracked until the end of 2011. The survival probability of IPOs is plotted against trading months, wherein the probability of survival of IPOs at that time is the percentage of cumulative survival at any given time. Further, the survival duration of IPOs determines the steepness of the curve. In order to show the sharpness of survival curve more clearly and closely to time, the plot of log survival function has been taken wherein the survival function is plotted on a logarithmic scale on the Y-axis (Garson, 2012).

In line with the findings of Boubakri *et al.* (2005), the survival function exhibits that the probability of surviving falls as the time from the issuance of IPO rises. A significant decline has been seen in this curve from zero to 50 months, which indicates a huge rate of non-survival during the initial years of IPOs. Thereafter, the rate of decline becomes moderate forming an elbow at around 50-60 months. This fall in survival function shows that chance of survival of IPOs in India is quite low during the first four to five years of issue i.e., during the hot issue period of 1992-1996. However, as the time increases, the rate of decline slows down and sustains around the probability value of 0.4 after 60 months of the issue.

*‘Insert Figure 2 here’*

Figure 3 exhibits the hazard function, which is exactly the opposite of survival function. It shows that the cumulative force of mortality of IPOs is very high in the first 50 months and reaches around the probability value of 0.6. Thereafter, an upward movement has been observed in the curve and the hazard probability remains closer to 0.9.

*‘Insert Figure 3 here’*

In other words, the hot issue period (1992-1996) brought several new issues in India, but most of such issues were ill fitted as they were of low quality and hence failed to survive longer in the aftermarket. This supports the ‘Window of Opportunity’ hypothesis which claims that most of the issues follow the herd behavior and enter the market in hot period, but in reality such issues are of low quality and hence they fail to fit in the rough market conditions (Raju and Prabhudesai, 2012; Demers and Joos, 2007; Kooli and Meknassi, 2007; Chi *et al.*, 2010).

**3.3 Survival Probability of IPOs:** The empirical investigation of the impact of market, along with offering and corporate variables on the survival probability of IPOs is conducted using logistic regression model. In this model, each IPO is tracked from the date of its listing until the next five years in order to classify it as ‘survivor’ or ‘non-survivor’. Majority of studies have tracked the status of IPOs for 5 years of listing as it is believed that five years is a sufficient period for analyzing their status in the market (Jain and Kini, 1999; Bhabra and Pettway, 2003; Peristiani and Hong, 2004; Demers and Joos, 2007; Howton, 2006; Chou *et al.*, 2007; Rath, 2008; Jain and Kini, 2008, to name a few). In addition, the period of five years covers major movements of a business cycle and is sufficient to examine the future prospects of an issue

(Rath, 2008). Moreover, the KM curves also exhibit that most of the IPOs in India failed to survive beyond this period. Hence, based upon this classification, out of 3125 IPOs from 1992 to 1996, 2258 are categorized as ‘survivors’ and 867 as ‘non-survivors’.

- The Model: Following logit model is estimated:

$$\begin{aligned} \text{Logit}(p) = & \alpha + \beta_1 \text{ Issue Size} + \beta_2 \text{ IPO Demand} + \beta_3 \text{ Initial Returns} \\ & + \beta_4 \text{ LM Reputation (n)} + \beta_5 \text{ LM reputation (size)} + \beta_5 \text{ Risk} \\ & + \beta_6 \text{ List Delay} + \beta_7 \text{ Market Level} + \beta_8 \text{ IPO Activity} + \beta_9 \text{ Age} \\ & + \beta_{10} \text{ Industry dummies} \end{aligned}$$

- Dependent Variable: The binary dependent variable in logit model takes the value ‘1’ if an IPO continues to survive for five years and ‘0’ if it gets delisted or stop trading from the exchange within this period.
- Explanatory Variables: As revealed from the literature, market factors are crucial in determining the success or failure of IPOs in the aftermarket. Hence, based upon the review, such factors are taken as explanatory variables in the model and their influence on the likelihood of survival is examined.
- Hypothesis: Based upon the intensive review of literature, the following hypothesis have been framed for logistic regression model:

H<sub>1a</sub>: Issues during the period of high market level are more likely to be delisted from the exchange.

H<sub>2a</sub>: Issues during the hot issue period (high IPO activity) are more likely to be delisted from the exchange.

The influence of market scenario is tested by taking two major variables i.e., Market level and IPO activity, but they also failed to exhibit any significant influence on the odds of survival of

IPOs in the aftermarket. This signifies that these market conditions do not determine the success or failure of IPOs within five year of listing. Further, the influence of other variables is also tested in the model. IPO demand, which represents the number of times an issue is subscribed, exhibits a positive and significant influence on the odds of survival of IPOs in the aftermarket. It clearly shows that the interest of investors towards an issue is crucial in determining its fitness on the exchange. The result corroborates with the findings of Kooli and Meknassi, 2007; Goot *et al.*, 2011; Raju and Prabhudesai, 2012, who also support the higher probability of survival of IPOs with higher demand. Further, lead manager's reputation (measured on the basis of size of issues they manage) exhibits a positive and significant influence on the survival prospects of IPOs. This finding supports that lead managers by the virtue of their expertise, reputed capital, wider network, and interlocking arrangements provide stronger support to the IPO firms that improves their survival profile in the aftermarket (Jain and Kini, 1999, Jain and Kini, 2000; Chou *et al.*, 2007; Kooli and Meknassi, 2007; Rath, 2008; Hamza and Kooli, 2010).

However, Initial returns, which refers to the returns on the first day of listing, exhibits a negative and significant influence on the post-listing status of IPOs. The rationale behind this negative impact is supported by several researchers who assert that issue with significant underpricing generates high indirect cost, less collected funds, and more financial difficulties on the firm which in turn decreases the likelihood of survival of its IPO in the aftermarket (Kooli and Meknassi, 2007; Hamza and Kooli, 2010; Raju and Prabhudesai, 2012). Similarly, 'List delay', i.e. the difference between the issue day and listing day, depicts a negative and significant influence on the odds of survival of IPOs in the aftermarket. The Indian primary market has faced a unique experience of a very long delay between the issue day and the listing day. Such delay is mainly due to time-consuming administrative procedure and postponement of the listing



day by the IPO issuing company. Hence, during this time lag, the market receives the sensitive information that may adversely affect the underpricing and initial volatility on the listing day (Shah, 1995; Chakrabarty and Ghosh, 2006). In other words, the issues with higher delay in listing exhibit lower chances of survival in the aftermarket. Hence, it is important that IPOs should be listed within the time limit as stipulated by SEBI otherwise it could prove to be detrimental for their survival on the exchange (Sehgal and Singh, 2008).

Age of the firm at the time of issue comes out to be one of the highly significant variables that positively influence the chance of survival of IPO in the aftermarket. It support that older firms by the virtue of their experience and wider knowledge about the market demonstrate a strong fit in the prevailing environment. However, the younger firms with a shorter operating history are more speculative and hence less likely to survive in the aftermarket (Hensler *et al.*, 1997; Peristiani and Hong, 2004; Audretsch and Lehmann, 2005; Demers and Joos, 2007; Carpentier and Suret, 2008; Chancharat *et al.*, 2008; Rath, 2008; Chi *et al.*, 2010). The odds ratio of age shows that each added year in a firm age leads to higher odds of survival of its IPO on the exchange. This clearly depicts that age of a firm at the time of its issue acts as a good predictor of the success of its IPO in the aftermarket.

The survival profile of IPOs varies across several industries as well. Taking manufacturing sector with the largest number of issues as a base, the model shows that out of all sectors, agriculture and administration sectors have a negative influence on the survival prospects of IPOs issued in these sectors. However, IPOs in mining, construction, wholesale and retail, accommodation, information and communication, finance and insurance, and others sectors have shown a higher likelihood of survival in the post-SEBI era. However, no significant influence is found for issue

size, risk, lead manager's reputation (measured as per number of issues they manage), transportation and storage sector in the model.

The goodness of fit of logit model is tested through 'Omnibus test of model coefficient' and 'Hosmer and Lemeshow'. The significant value of omnibus test and the insignificant value of Hosmer and Lemeshow indicate that the model has a good fit. The value of Nagelkerke R square comes out to be 13.3 per cent, which is quite closer to pseudo R square obtained by several researchers (Raju and Prabhudesai, 2012; Adjei *et al.*, 2008; Kooli and Meknassi, 2007; Chou *et al.*, 2007; Chi *et al.*, 2010). The overall classification percentage is 74.5 percent, which indicates that the model is quite good in predicting the correct category for survivors and non-survivors.

**3.4 Survival Duration of IPOs:** In order to explore the influence of market variables on the duration of IPOs in the aftermarket, the survival analysis methodology is best suited. Out of several models of survival analysis, the most efficient 'Accelerated Failure Time' (AFT) model is applied in order to check the robustness of results obtained from logistic regression (Hensler *et al.*, 1997; Jain and Kini, 2000; Kooli and Meknassi, 2007; Raju and Prabhudesai, 2012).

- **Functional Form of AFT model:** Since AFT is a parametric model, the baseline hazard function assumes to follow some distribution. There are several distribution forms of the AFT model such as Log-Normal, Log-Logistic, Exponential, Weibull and Gamma, out of which the best form is to be selected. In order to test the distribution form, the number of delisted and suspended companies is plotted against the time. Figure 4 shows that the peak of delisting reaches to its maximum and thereafter it slowly declines monotonically. This non-monotonic pattern in hazard suggests that log-normal or log-logistic functional forms are best suited for AFT model. Although both forms are quite similar in shapes, yet researchers support the log-

logistic over log-normal as it captures the censored data well and is not sensitive to smaller duration (Hensler *et al.*, 1997, Raju and Prabhudesai, 2012).

*‘Insert Figure 4 here’*

Another way of testing the distribution form is plotting the log-odds ratio of survival against the log survival time. As per Kleinbaum and Klein (2005), in order to assess the appropriateness of log-logistic assumption, the log-odds of survival should be a linear function with log of time with slope  $-\rho$ . Figure 5 shows the resultant output.

*‘Insert Figure 5 here’*

On the other hand, log-odds ratio of failure is also plotted against the log-survival time and is shown in figure 6.

*‘Insert Figure 6 here’*

Both the figures (5 & 6) shows a clear straight line, which validates that log-logistic is the best-fitted form of AFT model (Bradburn *et al.*, 2003; Kleinbaum and Klein, 2005, p. 279).

- **Log-Logistic AFT Model:** The log-logistic AFT model is estimated with the data previously described. In this, each of 3125 IPOs (1992-1996) are tracked from the date of their listing until the date of delisting or the end of 2011, whichever is earlier. This categorization resulted in 1450 survivors and 1675 non-survivors. Taking this data, the following model is estimated in which the natural logarithm of survival time is presented in linear function of explanatory variables:

$$\begin{aligned}
Ln(T_j) = & \alpha + \beta_1 \text{ Issue Size} + \beta_2 \text{ IPO Demand} + \beta_3 \text{ Initial Returns} \\
& + \beta_4 \text{ LM Reputation}(n) + \beta_5 \text{ LM reputation (size)} + \beta_5 \text{ Risk} \\
& + \beta_6 \text{ List Delay} + \beta_7 \text{ Market Level} + \beta_8 \text{ IPO Activity} + \beta_9 \text{ Age} \\
& + \beta_{10} \text{ Industry dummies}
\end{aligned}$$

- Dependent variable: In survival analysis, the dependent variable is the number of trading months of IPOs from the date of listing until the date of delisting or the end of 2011, whichever is earlier. Since the time window is different for each issue, the probability of survival or failure varies as per the length of post-issue period (Raju and Prabhudesai, 2012).
- Explanatory variables: The market level and IPO activity are the two main explanatory variables taken in the AFT model, along with other variables.
- Hypothesis: The log-logistic AFT model has been applied to test following hypothesis:  
H<sub>1b</sub>: Issues during the period of high market level survive for shorter duration in the aftermarket.  
H<sub>2b</sub>: Issues during the hot issue period (high IPO activity) survive for shorter duration in the aftermarket.

The results of AFT model for market related variables exhibit that out of both variables, IPO activity exhibits a negative and significant influence on the survival duration of IPOs in the aftermarket. This supports that hot issue period creates a conducive environment for the inferior firms to go for public issue, but such firms survive for shorter duration in the aftermarket (Demer and Joos, 2007; Kooli and Meknassi, 2007; Chi *et al.*, 2010; Raju and Prabhudesai, 2012). Hence, the results of market specific variables demonstrate the importance of market timings as well as the favorable market conditions for the longevity of IPOs on the trading exchange (Zhao, 2005).

Further, the model confirms that issue size, lead manager's reputation, IPO demand, and age of the firm accelerates the survival duration of IPOs, whereas, initial returns (MAER), risk, and list delay decelerates their duration on the trading exchange. Although the issue size variable was found to be insignificant in logit model (taking five year window) but is found to be highly significant in survival analysis model. Hence, this finding provides a strong support to the hypothesis and is in line with the liability of smallness theory of firm survival which asserts that large organizations have better prospects of survival than small firms (Aldrich, 1979; Bruderl *et al.*, 1992; Perez and Catillejo, 2008).

Similarly, it is found that lead manager's reputation (measured on the basis of size of issue managed) ensures the longer endurance of IPOs on the exchange. Also, the time ratio of IPO demand variable validates that when the demand for an issue increases, the survival time significantly accelerates in the aftermarket (Kooli and Meknassi, 2007; Goot *et al.*, 2011; Raju and Prabhudesai, 2012).

In line with the results of logit model, it is found that higher underpricing lowers the survival duration of IPOs in the marketplace. Also, the issues with higher listing delay fails to survive longer on the exchange. However, risk variable was found to be insignificant in logit model, but in AFT model this variable proves the hypothesis to be correct and support that that level of risk in the issue significantly lowers the survival duration of IPOs on the trading exchange (Bhabra and Pettway, 2003; Rath, 2008; Chi *et al.*, 2010; Goot *et al.*, 2011).

The positive influence of age on the survival duration goes in the line with the liability of newness theory of firm survival which suggests that since the new organization are highly dependent upon the cooperation of strangers and do not possess the ability to compete effectively

against the established firms, hence they fail to survive longer than older firms (Stinchcombe, 1965). Finally, it is obtained from the model that IPOs in agriculture and administration sectors survive for shorter duration, whereas IPOs in construction, wholesale and retail, accommodation, information and communication, finance and insurance, and others sectors survive longer in the aftermarket. However, no significant results are obtained from mining and transportation sector.

*'Insert Table 4 here'*

#### **4. CONCLUSION**

The success or failure of an IPO depends upon market conditions to a great extent. There are few researchers that have associated the market related factors with IPO survival (Hensler *et al.*, 1997; Kauffman and Wang, 2003; Demers and Joos, 2006). They claim that market related factors such as market activity, stock volatility, bond rating, etc. provide the benchmark in estimating the likelihood of survival or failure of IPO. Market conditions are most obvious and least controllable factors that influence the decision of a company to issue an IPO (Zhao, 2005). Such market conditions can change in a matter of days, sometimes even in hours. Hence, adequate preparation on the part of issuing firms can prove to be critical in this process (Rajan, and Servaes, 2002).

This paper explores the survival profile of initial public offerings in India during the most volatile market conditions, i.e. post-SEBI and hot-issue period of 1992-1996. The introduction of SEBI and the abolition of CCI brought the tremendous changes in the Indian primary market. During this fluctuative period, several firms followed the herd behavior and introduced their new issues in the market, however only a few managed to sustain their identity in the aftermarket. This phenomenon is provided in the 'Window of Opportunity' hypothesis or 'Hot issue'

phenomenon. The present study examines this theory in light of IPOs in the hot issue period by taking their market along with several issue and corporate specific characteristics. The sample of this study comprises of 3125 IPOs during 1992-1996 and they are tracked till the end of 2011. The empirical analysis has been done using most sophisticated methodologies i.e, Logistic Regression model, Kaplan-Meier estimation, and Log-Logistic Accelerated Failure Time (AFT) model.

The study validates the theory of 'Window of Opportunity' and exhibits that most of the issues in the hot period are of low quality who just entered the market to take the benefit of favorable market environment, but in reality they do not possess the ability to withstand the tough market conditions and hence they failed to survive in the aftermarket (Hensler *et al.*, 1997; Boubakri *et al.*, 2005; Demers and Joos, 2007). The analysis of market specific variables reveals that issues in the period of high IPO activity fail to sustain longer in the aftermarket. Further, the survival profile of IPOs across offering characteristics reveals that issue size, lead manager's reputation, and IPO demand have a positive influence, whereas initial returns, risk, and list delay have a negative influence on the endurance of IPOs on the trading exchange. The empirical analysis of age of the firm at the time of issue supports the hypothesis that older firms have more potential to sustain longer on the exchange as compared to younger firms. The survival profile of IPOs has been tested across the several industries as well, which exhibits that IPOs in agriculture and administration sectors have lower likelihood of survival and smaller survival duration, whereas IPOs in information and communication, construction, accommodation, wholesale and retail, finance and insurance, and other sectors have higher likelihood of survival and longer survival duration in the aftermarket. However, no significant influence could be obtained from mining and transport sectors. Overall, the KM estimation method shows a significant decline in the

survival rate and a growth in the hazard rate during the first 50-60 months of listing of IPOs on BSE.

The present study contributes to the IPO literature in general and survival in particular. Also, the findings of this study provide useful insight to the issuers, investors, regulators and policy makers. Issuers can take important decisions about the issue considering the long term prospects of IPO whereas investors can take the rational investment decisions based upon the chance of survival as well as their duration in the future (Ahmad, 2012). The regulators can formulate the stringent laws and policies so as to ensure a lucrative and more sustained market that will protect the interest of investors in the aftermarket.

## **REFERENCES**

- Adjei, F., Cyree, K.B. & Walker, M.M. (2008). The Determinants and Survival of Reverse Mergers Vs IPOs. *Journal of Economics and Finance*, 32(2), 176-194.
- Agarwal, R. & Gort, M. (2002). Firm and Product Life Cycle and Firm Survival. *American Economic Review*, 92(2), 184-190.
- Ahmad, W. (2012). Lockup Agreements and Survival of IPO firms. Proceedings of Annual conference of School of Economics and Business, University of Barcelona, Spain.
- Aldrich, H.E. (1979). Organizations and Environments. Prentice-Hall, Inc., Englewood Cliffs, New Jersey.
- Audretsch, D.B. & Lehmann, E.E. (2005). The Effects of Experience, Ownership and Knowledge on IPO Survival: Empirical Evidence from Germany. *Review of Accounting and Finance*, 4(4), 13-34.
- Audretsch, D.B. (1995). Innovation, Growth and Survival. *International Journal of Industrial Organization*, 13(4), 441-457.



Bhabra, H. & Pettway, R. (2003). IPO Prospectus Information and Subsequent Performance. *The Financial Review*, 38(3), 369-397.

Bhattacharya, U., Borisov, A. & Yu, X. (2011). Do Financial Intermediaries during IPOs affect Long-Term Mortality Rates?. <http://ssrn.com/abstract=1781242>. Accessed 17 December 2012.

Boubakri, N., Kooli, M. & L'Her, J.F. (2005). Is There Any Life After Going Public? Evidence from the Canadian Market. *The Journal of Private Equity*, 8(3), 30-40.

Bradburn, M.J, Clark, T.G, Love, S.B. & Altman, D.G. (2003). Survival analysis Part II: Multivariate Data Analysis-An Introduction to Concepts and Methods. *British Journal of Cancer*, 89(3), 431-436.

Bruderl, J., Preisendorfe, P. & Ziegler, R. (1992). Survival Chances of Newly Founded Business Organizations. *American Sociological review*, 57(2), 227-242.

Carpentier, C. & Suret, J.M. (2008). The Survival and Success of Canadian Penny Stock IPOs. *Small Business Economics*, 36(1), 101-121.

Chakrabarty, B. & Ghosh, S. (2006). Designing an Efficient IPO Mechanism: Evidence from E-IPOs. *Paper presented at Eight Annual Conference on Money, Banking and Finance* [http://www.igidr.ac.in/money/mfc\\_08/Designing%20an%20Efficient%20IPO..Bidisha%20&%20Saurabh%20Ghosh](http://www.igidr.ac.in/money/mfc_08/Designing%20an%20Efficient%20IPO..Bidisha%20&%20Saurabh%20Ghosh). Pdf. Accessed July 21 2014

Chancharat, N., Krishnamurti, C. & Tian, G. (2008). When the Going Gets Tough: Board Capital Structure and Survival of Newly Economy IPO Firms. *Paper presented at 21<sup>st</sup> Australian Finance and Banking Conference, Australia*.

Chi, J., McWha, M. & Young, M. (2010). The Performance and Survivorship of New Zealand IPOs. *International Review of Financial Analysis*, 9(3), 172-180.

Chou, T., Cheng, J. & Chien, C. (2007). Do Expert Intermediaries Improve the Survivability of IPOs?. [http://www.fin.ntu.edu.tw/~conference/conference2006/proceedings/proceeding/16/16-3\(A122\).pdf](http://www.fin.ntu.edu.tw/~conference/conference2006/proceedings/proceeding/16/16-3(A122).pdf). Accessed 4 April 2012.

Cockburn, I.M. & Wagner, S. (2007). Patents and the Survival of Internet Related IPOs. *Working Paper, National Bureau of Economic Research, Cambridge.*

Demers, E.A. & Joos, P. (2007). IPO Failure Risk. *Journal of Accounting Research*, 45(2), 333-371.

Donaldson, G. (1962). New Framework for Corporate Debt Policy. *Harvard Business Review*, 40 (March-April), 117-131.

Espenlaub, S., Goyal, A. & Mohamed, A. (2016). Impact of Legal Institutions on IPO Survival: A Global Perspective. *Journal of Financial Stability*, Forthcoming. <https://ssrn.com/abstract=2792221>. Accessed 13 August 2016,

Fama, E.F. & French, K.R. (2004). New Lists: Fundamentals and Survival Rates. *Journal of Financial Economics*, 73(2), 229-269.

Field, A. (2005). *Discovering Statistics Using SPSS*. Second edition, Sage publications, London.

Fischer, H.M. & Pollock, T.G. (2004). Effects of Social Capital and Power on Surviving Transformational Change: The Case of Initial Public Offerings. *Academy of Management Journal*, 47(4), 463-481.

Garson, G.D. (2012). Life tables & Kaplan-Meier Analysis. *Blue Book Series*, Statistical Associates Publishing, North Carolina State University, Australia.

Goot, T.V.D., Geirsbergen, N.V. & Botman, M. (2011). What Determine the Survival of Internet IPOs?. *Applied Economics*, 41(5), 547-561.

Hamza, O. & Kooli, M. (2010). Does Venture Capitalists Reputation Improve the Survival Profile of IPO Firms?. *Symposium EFM*, Canada.

Hensler, D.A., Rutherford, R.C. & Springer, T.M. (1997). The Survival of Initial Public Offerings in the Aftermarket. *The Journal of Financial Research*, 20(1) 1, 93-110.

Howton, S.W. (2006). The Effect of Governance Characteristics on the State of Firm after an Initial Public Offerings. *The Finance Review*, 41(3), 419-433.

- Ibbotson, R.G. & Jefferey, J. (1975). Hot Issue Markets. *Journal of Finance*, 30(4), 1027-1042.
- Jain, B.A & Kini, O. (1994). The Post-Issue Operating Performance of IPO Firms. *Journal of Finance*, 49(5), 1699-1726.
- Jain, B.A & Kini, O. (2008). Impact of Strategic Investment Choices on the Post-Issue Operating Performance and Survival of US IPO Firms. *Journal of Business Finance and Accounting*, 35(3-4), 459-490.
- Jain, B.A. & Kini, O. (1999). The Life Cycle of Initial Public Offering Firms. *Journal of Business Finance and Accounting*, 26(9-10), 1281-1307.
- Jain, B.A. & Kini, O. (2000). Does the Presence of Venture Capitalists Improve the Survival Profile of IPO Firms?. *Journal of Business Finance and Accounting*, 27(9-10), 1139-1183.
- Kaplan, E.L. & Meier, P. (1958). Non Parametric Estimation of Incomplete Observations. *Journal of the American Statistical Association*, 53(282), 457-481.
- Kleinbaum, D.B. & Klein, M. (2005). Survival analysis: A Self Learning Text. Second edition, Springer, US.
- Kooli, M. & Meknassi, S. (2007). The Survival Profile of US IPO Issuers 1985-2005. *The Journal of Wealth Management*, 10(2), 105-119.
- Lee, E.T. & Wang, J.W. (2003). Statistical Methods for Survival Data Analysis. Third edition, John Wiley & Sons, US.
- Li, D. & Lui, J. (2010). The Life Cycle of Initial Public Offering Companies: A Panel Analysis of Chinese Listed Companies. <http://www.ceauk.org.uk/2010-conference-papers/full-papers/Dairui-LI-CEA-final2.pdf>. Accessed 20 August 2013.
- Li, J., Zhang, L. & Zhou, J. (2006). Earnings Management and Delisting Risk of Initial Public Offerings. *Research Paper Series AAA 2008*, Financial Accounting and Reporting Section (FARS) Paper, Simon School, University of Rochester. <http://ssrn.com/abstract=641021>. Accessed 14 June 2013.

Liu, J. & Li, D. (2014). The life Cycle of Initial Public Offering Companies in China. *Journal of Applied Accounting Research*. 15. 291-307.

Loughran, T. & Ritter, J.R. (2004). Why has IPO Underpricing Changed Over Time?. *Financial Management*, 33(3), 5-37.

Lowery, T.J., Messina, F.M. & Richard, M.D. (1999). The Auditor's Going Concern Disclosure as a Self-Fulfilling Prophecy: A Discrete Time Survival Analysis. *Decision Sciences*, 30(3), 805-824.

Mills, M. (2010). *Fundamentals of Survival and Event History Analysis*. Sage Publications  
[http://www.sagepub.in/upm-data/39847\\_9781848601017\\_Mills.pdf](http://www.sagepub.in/upm-data/39847_9781848601017_Mills.pdf). Accessed 19 April 2013.

Noor, Z.M. & Iskandar, T.M. (2012). Corporate governance and Corporate failure: A Survival Analysis. *Prosing Perkem VII*, 1, 684-695.

Perez, S.E. & Castillejo, J.A. (2008). The Resource Based Theory of the Firm and Firm Survival. *Small Business Economics*, 30(3), 231-249.

Peristiani, S. & Hong, G. (2004). Pre-IPO Financial Performance and Aftermarket Survival. *Current Issues in Economics and Finance*, Federal Reserve Bank of New York, 10(2), 1-7.

Peristiani, S. (2003). Evaluating the Riskiness of Initial Public Offerings: 1980-2000. *Working Paper no. 167*, Federal Reserve Bank of New York.

Rajan, R. & Servaes, H. (2002). The Effect of Market Conditions on Initial Public Offerings. from <https://faculty.chicagobooth.edu/raghuram.rajan/research/papers/mktconditions.pdf> Accessed 17 December 2016.

Raju, G.A. & Prabhudesai, H. (2012). A Tale of Two Decades (1990–2010): The Survival Profile of Indian Initial Public Offering Issuers. *The Journal of Wealth Management*, 14(4), 109-121.

Rath, S. (2008). The Survival of Initial Public Offerings in Australia. *Paper presented at Oxford Business & Economics Conference program, UK*.  
[http://www.gcbe.us/2008\\_OBEC/data/confcd.htm](http://www.gcbe.us/2008_OBEC/data/confcd.htm). Accessed 30 August 2012.

Sehgal S. & Singh, B. (2008). Determinants of Initial and Long-Run Performance of IPOs in Indian Stock Market. *Asia-Pacific Business Review*, 4(4), 24-37.

Shah, A. (1995). The Indian IPO Market: Empirical Facts. Technical Report, Centre for Monitoring Indian Economy, Mimeo.

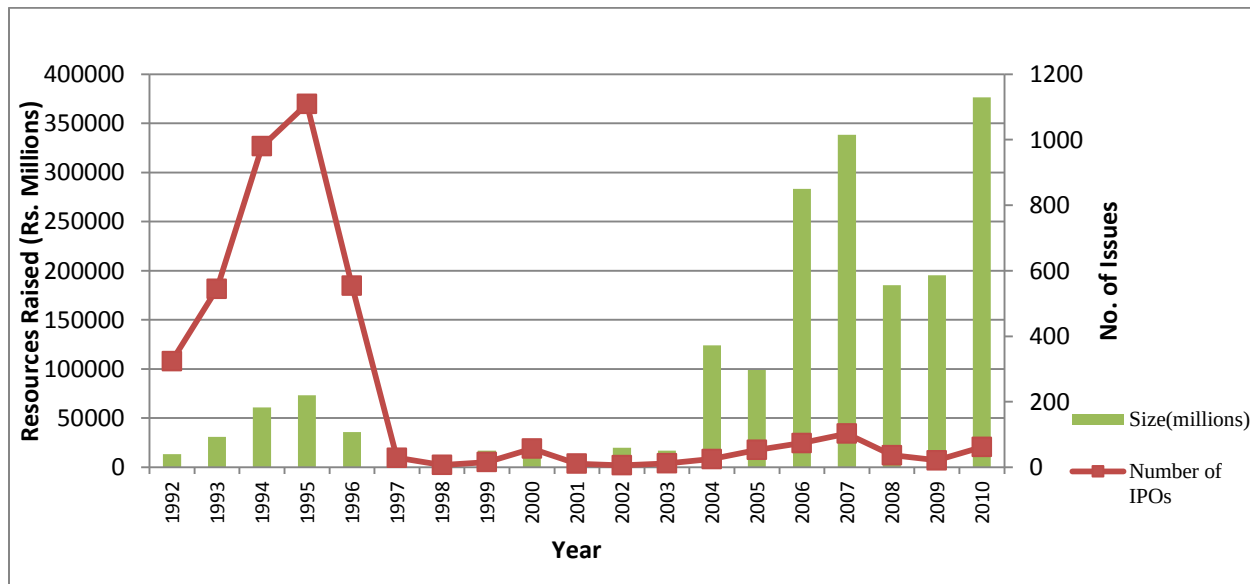
Stinchcombe, A.L. (1965). Social Structures and Organizations. *Advances in Strategic Management*, 17, 229-259.

Walter, J.E. (1957). Determination of Technical Insolvency. *Journal of Business*, 30(1), 30-43.

Zhao, X. (2005). What Causes Initial Public Offerings to be Unsuccessful? An Empirical Analysis. PhD thesis, Mississippi State University, Mississippi.

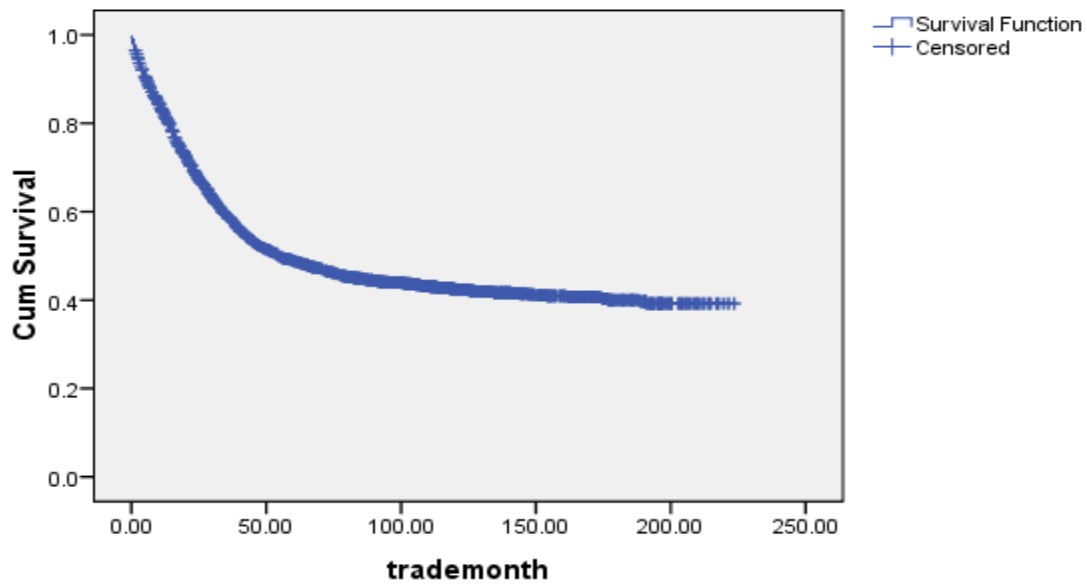
Zingales, L. (1995). Insider Ownership and the Decision to Go Public. *Review of Economics Studies*, 62(3), 425-448.

**Figure 1: Trends in Indian IPO Market**



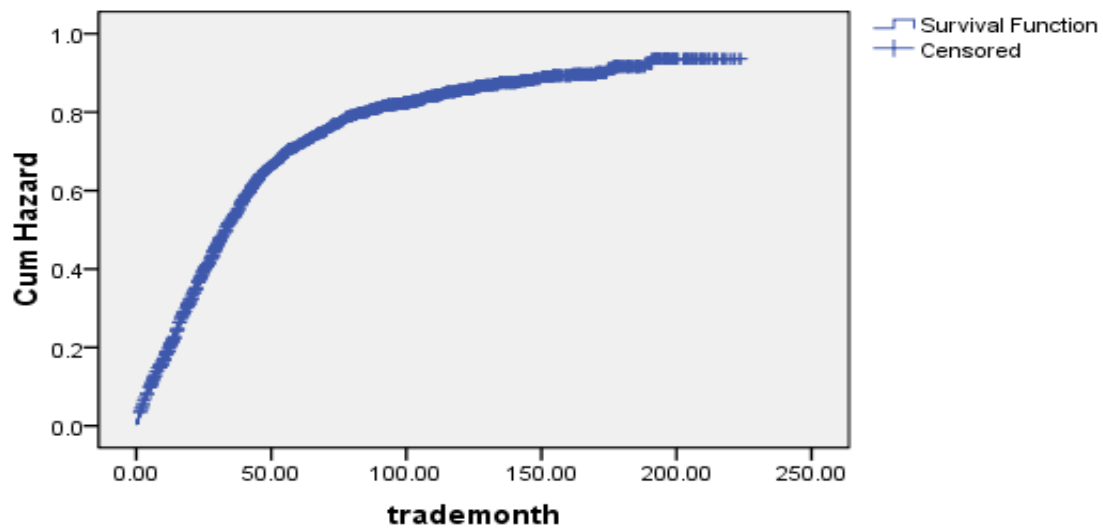
Source: Prime Database

**Figure 2: Survival Function of IPOs**



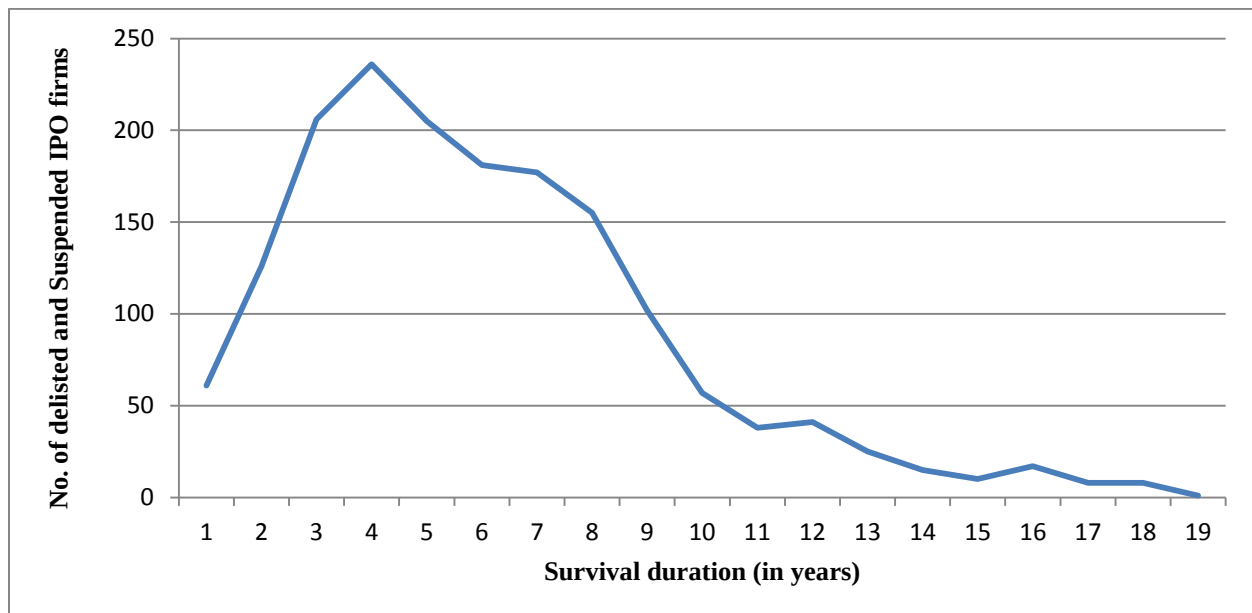
Source: Stata

**Figure 3: Hazard Function of IPOs**



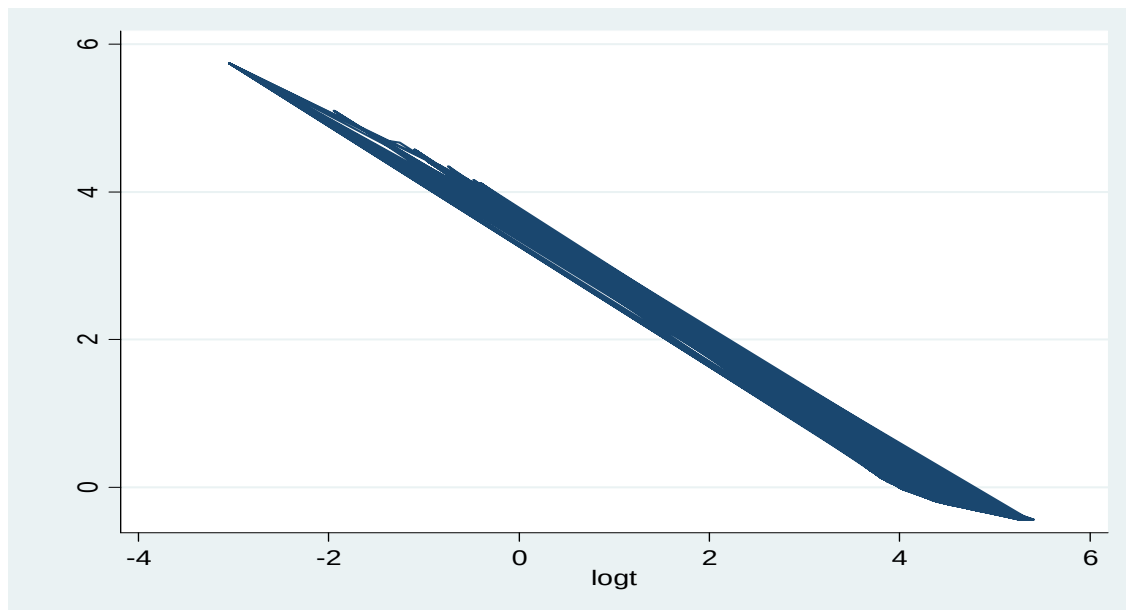
Source: Stata

**Figure 4: Delisting Frequency Distribution**



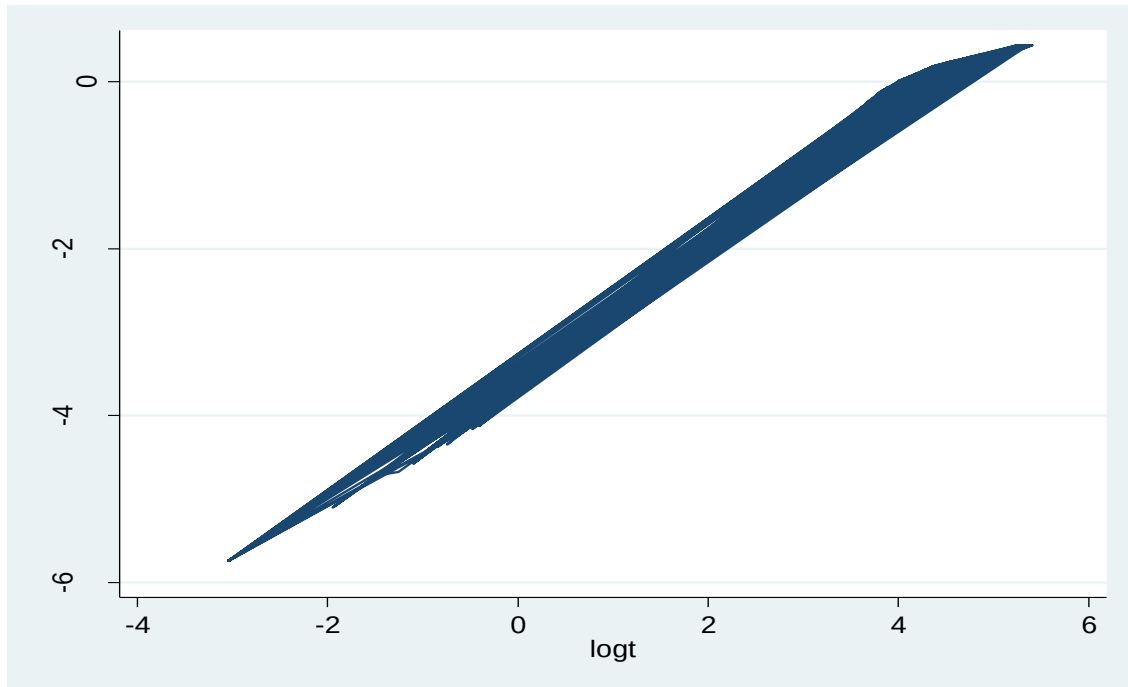
*Source: Author's calculation*

**Figure 5: Survival Plot**



*Source: Stata*

**Figure 6: Failure Plot**



*Source: Stata*



**Table 1: Measurement of Variables**

| <b>Variable</b>                               | <b>Variable defined</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>Expected relationship to survival</b> |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <b>Market Variables</b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          |
| <b>Market Level</b>                           | Return on Sensex for the month of issue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -                                        |
| <b>IPO Activity</b>                           | The natural logarithm of the number of issues in the calendar quarter of the offering.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | -                                        |
| <b>Other Variables</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          |
| <b>Issue Size</b>                             | The natural logarithm of the size of the offering listed in the prospectus, or the amount raised by the company in the issue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | +                                        |
| <b>Lead manager's reputation</b>              | <p>Meggison and Weiss (1991) reputation measure based upon a number of issues and total size of issues managed.</p> <ul style="list-style-type: none"> <li>On the basis of number of issues managed by lead managers:<br/>LM Reputation (n)= Percentage of number of issues managed by lead managers i.e. Total number of issues managed by LM/ Total number of issues in the sample</li> <li>On the basis of size of issues managed by lead managers:<br/>LM Reputation (size)= Percentage of total issue size managed by lead managers i.e. Total issues size managed by LM/ Total issue size of all the issues in the sample</li> </ul> <p>In case an issue has more than one lead manager, the average of lead manager's share is used as a measure of quality (Meggison and Weiss, 1990, p.13).</p> | +                                        |
| <b>Initial Returns (MAER or underpricing)</b> | <p>Raw returns= (Closing price on the listing day– Offering price) / (Offering price)</p> <p>Market returns= Closing value of Sensex on listing date- Closing value of Sensex on Issue date/ Closing value of Sensex on Issue date</p> <p>Market adjusted excess returns (MAER) = Raw returns- Market returns</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | +/-                                      |
| <b>IPO demand</b>                             | The natural logarithm of the number of times issue has been subscribed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | +                                        |
| <b>Risk</b>                                   | Standard deviation of first 30 trading days of aftermarket returns (Jain and Kini, 1999)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -                                        |
| <b>List delay</b>                             | The natural logarithm of the difference between Issue date and List date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | -                                        |
| <b>Age of Company</b>                         | The natural logarithm of the one plus the difference between incorporation year and the year of issue.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | +                                        |
| <b>Industry</b>                               | Binary industry dummies based upon NIC 2008 classification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | +/-                                      |

Source: Compiled from various studies

**Table 2: IPO characteristics across survivors and non-survivors**

| Variables                             | Survivor IPOs (2258) |        |                | Non-Survivor IPOs (867) |        |                | T value  | Wilcoxon Z value |
|---------------------------------------|----------------------|--------|----------------|-------------------------|--------|----------------|----------|------------------|
|                                       | Mean                 | Median | Std. Deviation | Mean                    | Median | Std. Deviation |          |                  |
| Market Variables                      |                      |        |                |                         |        |                |          |                  |
| Market Level (Percentage)             | 0.44                 | -1.30  | 8.86           | 0.90                    | -0.68  | 8.60           | -1.302   | -1.86*           |
| IPO Activity (No. of issues)          | 237.84               | 231.00 | 108.47         | 230.00                  | 230.00 | 95.44          | 1.868*   | -1.08            |
| Other Variables                       |                      |        |                |                         |        |                |          |                  |
| Issue size (Crores)                   | 6.86                 | 3.00   | 41.76          | 4.22                    | 3.16   | 4.87           | 1.856*   | -1.23            |
| Lead manager rep. (n) (Percentage)    | 1.70                 | 1.13   | 1.61           | 1.60                    | 0.93   | 1.61           | 1.526    | -2.35***         |
| Lead manager rep. (size) (Percentage) | 1.47                 | 0.23   | 2.70           | 0.92                    | 0.18   | 1.90           | 5.511*** | -4.50***         |
| Initial Returns (Percentage)          | 43.59                | 20.12  | 129.53         | 45.83                   | 19.69  | 109.45         | -0.451   | -1.46            |
| IPO Demand (No. of times)             | 7.96                 | 2.19   | 22.88          | 4.79                    | 1.21   | 9.51           | 3.962*** | -9.84***         |
| Risk (Percentage)                     | 0.13                 | 0.09   | 0.43           | 0.13                    | 0.11   | 0.13           | -0.352   | -6.62***         |
| List Delay (Days)                     | 133.22               | 82.00  | 240.62         | 148.70                  | 87.00  | 184.80         | -1.710*  | -5.49***         |
| Age (Years)                           | 7.31                 | 5.00   | 7.64           | 6.14                    | 5.00   | 5.89           | 4.076*** | -3.11***         |

Note: \*\*\*Significant at 1% level, \*\* Significant at 5% level, \* Significant at 10% level

**Table 3: Means and Medians for Survival Time**

| Mean     |                         |             | Median   |                         |             |
|----------|-------------------------|-------------|----------|-------------------------|-------------|
| Estimate | 95% Confidence Interval |             | Estimate | 95% Confidence Interval |             |
|          | Lower Bound             | Upper Bound |          | Lower Bound             | Upper Bound |
| 108.924  | 105.274                 | 112.573     | 54.619   | 47.136                  | 62.102      |

**Table 4: Results of Logistic Regression Model and Survival Analysis**

| Independent variables                       | Logistic Regression Model |      |           |             | Log-Logistic AFT model |         |          |                           |
|---------------------------------------------|---------------------------|------|-----------|-------------|------------------------|---------|----------|---------------------------|
|                                             | $\beta$                   | SE   | Wald      | Exp $\beta$ | B                      | SE      | Wald     | Time Ratio (Exp $\beta$ ) |
| <b>Market Variables</b>                     |                           |      |           |             |                        |         |          |                           |
| Market level (Percentage)                   | -.005                     | .005 | 1.184     | .995        | -0.0065                | 0.0042  | -1.54    | 0.993                     |
| IPO Activity (Number of issues)             | -.133                     | .095 | 1.952     | .875        | -0.161                 | 0.0853  | -1.89*   | 0.851                     |
| <b>Control Variables</b>                    |                           |      |           |             |                        |         |          |                           |
| Issue size (Crore)                          | -.033                     | .067 | .251      | .967        | 0.125                  | 0.056   | 2.23**   | 0.882                     |
| Lead Manager's reputation_size (Percentage) | .106                      | .027 | 15.494*** | 1.111       | 0.089                  | 0.020   | 4.47***  | 1.093                     |
| Lead Manager's reputation_n (Percentage)    | -.018                     | .035 | .248      | .983        | 0.005                  | 0.029   | 0.20     | 1.005                     |
| Initial Returns (Percentage)                | -.002                     | .000 | 18.426*** | .998        | -0.0019                | 0.00035 | -5.54*** | 0.998                     |
| IPO demand (No. of times)                   | .371                      | .043 | 74.045*** | 1.450       | 0.529                  | 0.369   | 14.33*** | 1.697                     |
| Risk (Percentage)                           | .096                      | .142 | .454      | 1.100       | -0.281                 | 0.0926  | -3.03*** | 0.755                     |
| List delay (No. of days)                    | -.301                     | .081 | 13.740*** | .740        | -0.337                 | 0.0797  | -4.24*** | 0.713                     |
| Age (Years)                                 | .027                      | .007 | 13.005*** | 1.027       | 0.145                  | 0.0513  | 2.83***  | 1.156                     |

| Industry Dummies                                           |       |      |                |        |                          |               |          |         |
|------------------------------------------------------------|-------|------|----------------|--------|--------------------------|---------------|----------|---------|
| Agriculture                                                | -.951 | .218 | 19.052***      | .386   | -1.045                   | 0.194         | -5.38*** | 0.351   |
| Mining                                                     | .639  | .346 | 3.413*         | 1.895  | 0.114                    | 0.279         | 0.41     | 1.120   |
| Construction                                               | .718  | .266 | 7.293***       | 2.050  | 1.277                    | 0.248         | 5.13***  | 3.585   |
| Wholesale and retail                                       | .831  | .212 | 15.289***      | 2.295  | 0.409                    | 0.167         | 2.44**   | 1.505   |
| Transport and storage                                      | .311  | .444 | .490           | 1.365  | 0.285                    | 0.385         | 0.74     | 1.329   |
| Accommodation                                              | .853  | .456 | 3.504*         | 2.346  | 1.04                     | 0.386         | 2.69***  | 2.829   |
| Information and communication                              | 2.538 | .404 | 39.521***      | 12.652 | 1.988                    | 0.209         | 9.50***  | 7.300   |
| Finance and insurance                                      | .330  | .125 | 6.895**        | 1.390  | 0.258                    | 0.117         | 2.20**   | 1.294   |
| Administration and support services                        | -.996 | .310 | 10.359***      | .369   | -0.762                   | 0.285         | -2.67*** | 0.466   |
| Others                                                     | 1.061 | .338 | 9.878***       | 2.890  | 1.175                    | 0.290         | 4.04***  | 3.238   |
| Constant                                                   | 2.388 | .775 | 9.504***       | 10.896 | 5.500                    | 0.72289       | 7.55***  | 244.691 |
| Omnibus Tests of Model Coefficients (Chi square (p value)) |       |      | 302.356(0.000) |        | Log Likelihood           | -4482.259     |          |         |
| Nagelkerke R Square                                        |       |      | 0.133          |        |                          |               |          |         |
| Hosmer and Lemeshow(Chi square (p value))                  |       |      | 4.320(0.827)   |        | LR (Chi square, p value) | 522.45(0.000) |          |         |
| Overall Classification percentage                          |       |      | 72.5%          |        |                          |               |          |         |

Note: \*\*\*Significant at 1% level, \*\* Significant at 5% level, \* Significant at 10% level;  $\beta$  =Beta; SE=Standard Error; Exp  $\beta$  stands for Exponential Beta

